BioSync

Translating Biological Signals into Robotic Control — A Proof-of-Concept, low-cost human-machine interface prototype

Purpose

BioSync investigates whether low-cost consumer hardware, such as webcams, can replace expensive biometric sensors for controlling robotic systems using biological inputs like eye movement, hand gestures, and facial motion.

Industrial Biometric trackers/scanners:

Current Disadvantages:	Current Advantages:
1.-Expensive Technology 2.- Not accessible to everyone 3.-Not Open Source 4.- Needs time to produce and distribute in physical form	1.- High Security and Protection 2.- Speed & Accuracy 3.- Durability & Stability 4.- Specialized, accurate Technology

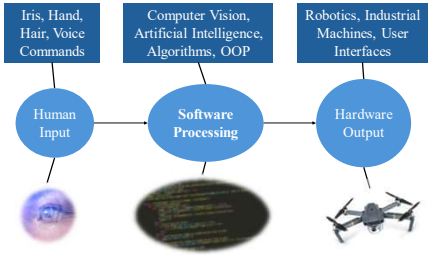
Alternative Explored:

Current Disadvantages:	Current Advantages:
1.- No Security or Protection 2.- Expected Jitters and Failures 3.- Still in development, non-specialized Technology 4.- Breaks under different Webcam frame ratios	1.- Uses Widely available Consumer Hardware such as WebCams 2. Accessible to anyone with a device with a camera integrated 3.- Open Source for Research, Investigation, and improvement. 4.- Instant, Digital Software

Background Research

Human–Machine Interfaces allow biological signals to control machines. Modern systems often require specialized hardware, increasing cost and limiting accessibility. Advances in computer vision tools like MediaPipe and OpenCV enable biological tracking using standard cameras. Combining these technologies with wireless microcontrollers allows affordable robotic control systems.

Human-Machine Interface:



Engineering Goals

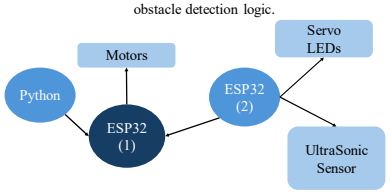
1.- Develop and refine a biological input robotic control prototype

2.- Demonstrate low-cost human–machine interface feasibility

3.- Evaluate system performance, limitations, and reliability

System Overview

BioSync uses computer vision software to detect biological inputs and transmit control commands wirelessly to a robotic platform. A dual-controller ESP32 system separates movement control from safety and obstacle detection logic.



Software

Modular Data Manipulation

BioSync organizes computer vision landmark data using structured dictionaries instead of indexed lists. This improves debugging, scalability, and gesture tracking accuracy. Modular visualization tools assist in validating gesture detection during development.

Original (get X values off the tip of your thumb):

```
tip of thumb = landmarkList[4][1]
```

Original (Trace your thumb if needed):

```
cv2.circle(flipCam,(center_x_fthumb,center_y_fthumb),size,color,cv2.FILLED)
cv2.circle(flipCam,(center_x_fthumbMiddle,center_y_fthumbMiddle),size,color,cv2.FILLED)
cv2.circle(flipCam,(center_x_fthumbTip,center_y_fthumbTip),size,color,cv2.FILLED)
```

After Modular Organization (Nested Dictionaries):

```
tip of thumb = landmarkList[fingers['Thumb']][1]
```

After Modular Organization (Functions with Default Parameters):

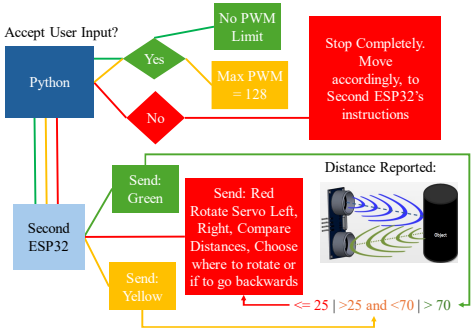
```
fingerTracing(color = (255,0,0),size = 12,thumbFinger=True)
```

Confusing Harder Debugging Time Consuming → Modular Easy-to-read code Intuitive

Landmark Reference:

Obstacle Avoidance

BioSync includes an autonomous obstacle avoidance system using ultrasonic sensing and servo-based environmental scanning. A three-state, 270 degree Environmental scan safety protocol, adjusts speed or overrides user commands to prevent collisions.



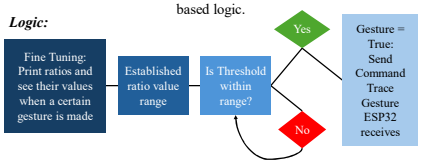
Limitations

Instance:	Limitation:
Continous Command Transmission	Reduces Network Efficiency
Wireless Communication	Environmental Obstruction Delay
Gesture Detection	Sensitive to lighting and positioning
Power Consumption	Doesn't allow for Extensive Operation
UltraSonic Sensing	Introduces delay at High Speeds

Gesture/Direction Recognition & Processing

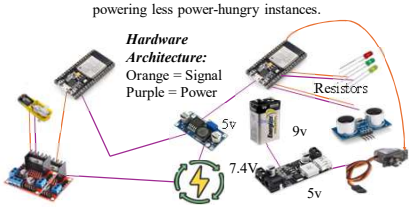
BioSync identifies biological inputs by analyzing landmark coordinate data obtained from computer vision tracking. Each detected landmark contains x and y positional values, which are used to determine user gestures and eye direction through mathematical comparisons (ratios, distances, etc) and threshold-based logic.

Logic:



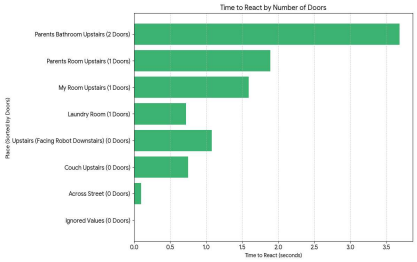
Hardware

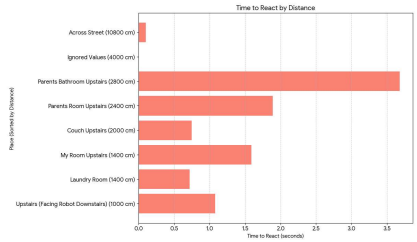
BioSync hardware evolved through iterative testing to improve durability, mobility, and reliability. Upgrades included ESP32 microcontrollers, mecanum wheels, high-torque motors, improved servo systems, and lithium-ion battery power, with 9V battery powering less power-hungry instances.



Testing & Data

Experiment:
Push the limits of wireless communication across distances, obstacles, and places.



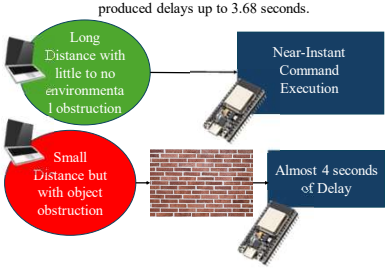


Testing & Data

Place	Distance (cm)	Doors	Curves	Altitude	Time to React
Ignored Values (Reset to Robot, 500, 1000, 4000 CM away Linear)	20,100,400,1000,4000...	0	0	0	0
Across Street	10800	0	0	0	0.10-0
Laundry Room	1400	1	2	0	0.72
Upstairs (Facing Robot Downstairs)	1000	0	0	2ft	1.08
My Room Upstairs	1400	1	1	2ft	1.59
Couch Upstairs	2000	0	1	2ft	0.75
Parents Room Upstairs	2400	1	2	2ft	1.89
Parents Bathroom Upstairs	2800	2	5	2ft	3.68

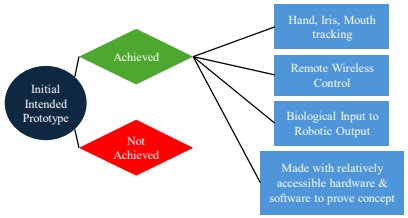
Results

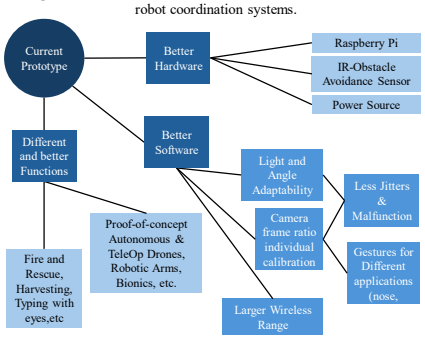
Wireless latency testing showed that environmental barriers significantly increased response delay. Open environments produced near-instant control, while obstructed environments produced delays up to 3.68 seconds.



Conclusion

BioSync demonstrates that biological inputs can reliably control robotic systems using low-cost computer vision and wireless communication technologies. Iterative software improvements increased gesture stability and system reliability, supporting future real-world applications.



Future Applications

Future development focuses on improving system modularity, wireless reliability, and AI processing capability. BioSync may expand into drone control, assistive robotics, bionics, and multi-robot coordination systems.